

ECE 332 — Homework 3: Generated Answers from Final Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Problem 1 — LHC Wave Normally Incident on a Dielectric

20 pts

Setup: 200 MHz LHC plane wave in air, $|E| = 5$ V/m, normally incident on dielectric ($\epsilon_r = 4$) at $z \geq 0$. Positive max at $z = 0$, $t = 0$.

NORMAL INCIDENCE — Γ AND τ

Reflection coeff. *boundary at $z = 0$, wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Lossless η_i $\sigma = 0$

$$\eta_i = \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Low-loss η_2 (poor conductor) $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \text{ (}\Omega\text{)}$$

For boundary Γ/τ , take just $|\eta_2| \approx \eta_0/\sqrt{\epsilon_{r2}}$; the j term is tiny.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

$\tilde{\mathbf{E}}$ PHASOR TEMPLATES

Wave in $+\hat{z}$, incident has unit-vector pol. $\hat{e}$, amplitude a_0 :

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 z}$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{+jk_1 z} \text{ (travels in } -\hat{z}\text{)}$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-jk_2 z} \text{ (lossless medium 2)}$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-\alpha_2 z} e^{-j\beta_2 z} \text{ (lossy medium 2)}$$

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

Wave parameters in medium 1 (air):

Angular frequency from f :

$$\omega = 2\pi f = 2\pi(2 \times 10^8) = 4\pi \times 10^8 \text{ rad/s}$$

Wavenumber from the lossless plane wave formula $k = \omega\sqrt{\epsilon_r}/c$. In air, $\epsilon_r = 1$ so this reduces to $k_1 = \omega/c$:

$$k_1 = \frac{\omega}{c} = \frac{4\pi \times 10^8}{3 \times 10^8} = \frac{4\pi}{3} \text{ rad/m}$$

(a) **LHC incident phasor.** The PHASOR TEMPLATES section says the incident wave has the form $\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 z}$ where $\hat{e}$ depends on polarization. From the polarization table:

$$\text{LHC} \Rightarrow \hat{e} = \hat{x} + j\hat{y}$$

With $a_0 = 5$ V/m (the modulus) and $k_1 = 4\pi/3$:

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

(b) Reflection coefficient Γ and transmission coefficient τ .

General intrinsic impedance: $\eta = \sqrt{\mu/\varepsilon}$.

Decompose into relative + free-space constants: since $\mu = \mu_r \mu_0$ and $\varepsilon = \varepsilon_r \varepsilon_0$,

$$\eta = \sqrt{\frac{\mu_r \mu_0}{\varepsilon_r \varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \cdot \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \eta_0$$

which for nonmagnetic media ($\mu_r = 1$) reduces to $\eta = \eta_0 / \sqrt{\varepsilon_r}$.

Compute η_0 from μ_0, ε_0 directly:

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{(4\pi \times 10^{-7})(36\pi \times 10^9)} = 120\pi \Omega \approx 377 \Omega$$

Medium 1 (air): $\mu_r = 1, \varepsilon_r = 1$. So $\eta_1 = \eta_0 \sqrt{1/1} = \eta_0 = 120\pi \Omega$.

Medium 2 (nonmagnetic dielectric, $\mu_r = 1, \varepsilon_r = 4$): use the shortcut $\eta = \eta_0 / \sqrt{\varepsilon_r}$:

$$\eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_{r2}}} = \frac{120\pi}{\sqrt{4}} = \frac{120\pi}{2} = 60\pi \Omega$$

Now apply the boundary formula $\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$:

$$\Gamma = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \frac{-60\pi}{180\pi} = \boxed{-\frac{1}{3}}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

(c) Reflected, transmitted, and total field phasors.

Wavenumber in medium 2, again from $k = \omega \sqrt{\varepsilon_r} / c$ but now $\varepsilon_r = 4$:

$$k_2 = \frac{\omega \sqrt{4}}{c} = 2k_1 = \frac{8\pi}{3} \text{ rad/m}$$

From the PHASOR TEMPLATES (reflected wave travels in $-\hat{z}$, sign of z -term flips):

$$\tilde{\mathbf{E}}^r = \Gamma \cdot a_0 \hat{e}^{+jk_1 z} = -\frac{1}{3} \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{+j4\pi z/3}$$

Transmitted wave in lossless medium 2:

$$\tilde{\mathbf{E}}^t = \tau \cdot a_0 \hat{e}^{-jk_2 z} = \frac{2}{3} \cdot 5(\hat{x} + j\hat{y}) e^{-j8\pi z/3} = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j8\pi z/3}$$

Total field in medium 1 is incident + reflected:

$$\boxed{\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x} + j\hat{y}) \left(e^{-j4\pi z/3} - \frac{1}{3} e^{+j4\pi z/3} \right) \text{ V/m}}$$

(d) Percentages of incident average power reflected and transmitted.

From the POWER REFLECTION snippet, $\%P_r = 100|\Gamma|^2$ and $\%P_t = 100|\tau|^2(\eta_1/\eta_2)$. The η_1/η_2 factor accounts for the change in wave impedance: even though $|\tau| > 0$, the medium beyond carries power differently.

$$\%P_r = 100 \cdot \left| -\frac{1}{3} \right|^2 = 100 \cdot \frac{1}{9} = \boxed{11.1\%}$$

$$\%P_t = 100 \cdot \left| \frac{2}{3} \right|^2 \cdot \frac{120\pi}{60\pi} = 100 \cdot \frac{4}{9} \cdot 2 = \boxed{88.9\%}$$

Sanity check: $11.1\% + 88.9\% = 100\%$. Energy conserved.

Problem 2 — Same Wave, Poor-Conductor Medium 2

20 pts

Setup: Replace medium 2 with poor conductor: $\epsilon_r = 2.25$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m. Everything else same as P1.

NORMAL INCIDENCE — Γ AND τ

Reflection coeff. *boundary at $z = 0$, wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Lossless η_i $\sigma = 0$

$$\eta_i = \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \text{ }\Omega$$

Low-loss η_2 (poor conductor) $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'} \right) \text{ (}\Omega\text{)}$$

For boundary Γ/τ , take just $|\eta_2| \approx \eta_0/\sqrt{\epsilon_{r2}}$; the j term is tiny.

LOW-LOSS QUICK PARAMS (med 2)

Check first $\sigma/(\omega\epsilon)$

$$\alpha_2 \approx \frac{\sigma_2}{2} \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r2}}} \text{ (Np/m)}$$

$$\beta_2 \approx \omega \sqrt{\mu_2 \epsilon_2} = \frac{\omega \sqrt{\epsilon_{r2}}}{c} \text{ (rad/m)}$$

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r2}}} \angle \approx 0$$

$$\epsilon = \epsilon_r \epsilon_0; \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}; \quad 36\pi\epsilon_0 \approx 1 \times 10^{-9}.$$

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

(a) Medium 1 (air) is unchanged from P1. Same incident phasor:

$$\eta_1 = 120\pi \text{ }\Omega, \quad k_1 = \frac{4\pi}{3} \text{ rad/m}$$

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

(b) Classify medium 2 first. The LOSSY classification needs the loss tangent $\sigma/(\omega\epsilon)$:

$$\frac{\sigma_2}{\omega\epsilon_2} = \frac{\sigma_2}{\omega\epsilon_r\epsilon_0} = \frac{10^{-4}}{(4\pi \times 10^8)(2.25)(8.854 \times 10^{-12})} = 4 \times 10^{-3}$$

This is $\ll 1$, so medium 2 is a **low-loss dielectric**. Use the LOW-LOSS QUICK PARAMS approximations, not the full lossy formulas.

Attenuation constant (small but nonzero now):

$$\alpha_2 \approx \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r2}}} = \frac{10^{-4} \cdot 120\pi}{2\sqrt{2.25}} = \frac{0.0377}{3} = 1.26 \times 10^{-2} \text{ Np/m}$$

Phase constant (same as lossless approximation since σ is tiny):

$$\beta_2 \approx \frac{\omega \sqrt{\epsilon_{r2}}}{c} = \frac{(4\pi \times 10^8)(1.5)}{3 \times 10^8} = 2\pi \text{ rad/m}$$

Impedance. Same decomposition $\eta = \sqrt{\mu_r/\epsilon_r} \eta_0$ as in P1, with the low-loss approximation $\eta_c \approx \eta_0/\sqrt{\epsilon_r}$ since the $j\sigma/(2\omega\epsilon')$ correction is negligible here ($\sigma/\omega\epsilon = 4 \times 10^{-3}$):

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{\sqrt{2.25}} = \frac{120\pi}{1.5} = 80\pi \text{ }\Omega$$

Now apply the boundary formulas exactly as in P1:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \frac{-40\pi}{200\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1 + \Gamma = \boxed{\frac{4}{5}}$$

(c) **Field phasors.** Same templates as P1 except the transmitted wave now has the lossy decay factor $e^{-\alpha_2 z}$:

$$\tilde{\mathbf{E}}^r = \Gamma \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -(\hat{x} + j\hat{y}) e^{+j4\pi z/3} \text{ V/m}$$

$$\tilde{\mathbf{E}}^t = \tau \cdot 5(\hat{x} + j\hat{y}) e^{-\alpha_2 z} e^{-j\beta_2 z} = 4(\hat{x} + j\hat{y}) e^{-1.26 \times 10^{-2} z} e^{-j2\pi z} \text{ V/m}$$

$$\boxed{\tilde{\mathbf{E}}_1 = 5(\hat{x} + j\hat{y}) \left(e^{-j4\pi z/3} - \frac{1}{5} e^{+j4\pi z/3} \right) \text{ V/m}}$$

(d) **Power percentages.** Same formulas as P1:

$$\%P_r = 100|\Gamma|^2 = 100 \cdot \frac{1}{25} = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100 \cdot \frac{16}{25} \cdot \frac{120\pi}{80\pi} = 100 \cdot \frac{16}{25} \cdot \frac{3}{2} = \boxed{96\%}$$

Sanity check: $4\% + 96\% = 100\%$. The dielectric is denser than air ($\epsilon_r = 2.25$ vs 1), so most of the wave gets through.

Problem 3 — Parallel-Polarized Wave Through Two Dielectric Layers 10 pts

Setup: $\theta_i = 30^\circ$ from air, layer 1 ($\varepsilon_r = 6.25$, $t = 5$ cm), layer 2 ($\varepsilon_r = 2.25$, $t = 5$ cm), exits to air.

SNELL'S LAW — OBLIQUE INCIDENCE	LATERAL DISPLACEMENT												
Single boundary <i>angles from normal</i> $\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$ Multilayer chain <i>stack of slabs 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4</i> $\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$ Air-slabs-air shortcut <i>outer media identical</i> $\theta_{\text{exit}} = \theta_{\text{incident}}$ <i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	Beam offset through N slabs <i>thickness t_i, refracted angle θ_i</i> $d = \sum_{i=1}^N t_i \tan \theta_i$ <i>Each slab adds $t_i \tan \theta_i$. Final air-to-air exit returns to θ_1 but is laterally shifted by d.</i> Common values <i>arctan</i> <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr> <th style="padding: 2px 5px;">sin θ</th> <th style="padding: 2px 5px;">θ</th> </tr> </thead> <tbody> <tr><td style="text-align: center;">0.2</td><td style="text-align: center;">11.54°</td></tr> <tr><td style="text-align: center;">1/3</td><td style="text-align: center;">19.47°</td></tr> <tr><td style="text-align: center;">1/2</td><td style="text-align: center;">30°</td></tr> <tr><td style="text-align: center;">0.5774</td><td style="text-align: center;">35.26°</td></tr> <tr><td style="text-align: center;">1/√2</td><td style="text-align: center;">45°</td></tr> </tbody> </table>	sin θ	θ	0.2	11.54°	1/3	19.47°	1/2	30°	0.5774	35.26°	1/√2	45°
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(a) Chained Snell's law:

$$\sin \theta_2 = \sin 30^\circ \sqrt{\frac{1}{6.25}} = 0.5 \cdot 0.4 = 0.2 \Rightarrow \theta_2 = \sin^{-1}(0.2) = \boxed{11.54^\circ}$$

$$\sin \theta_3 = 0.2 \sqrt{\frac{6.25}{2.25}} = 0.2 \cdot \frac{5}{3} = \frac{1}{3} \Rightarrow \theta_3 = \sin^{-1}(1/3) = \boxed{19.47^\circ}$$

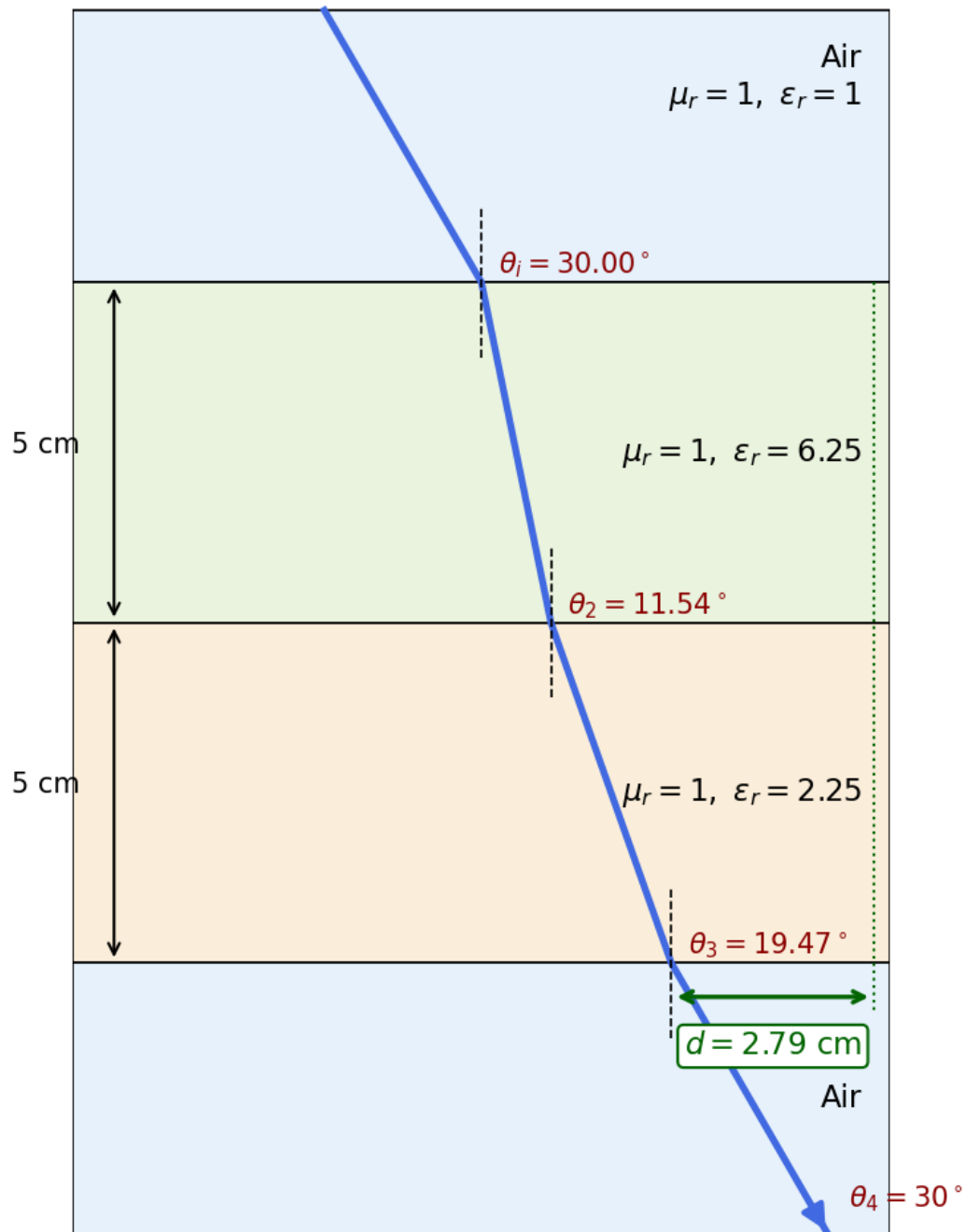
$$\sin \theta_4 = \frac{1}{3} \sqrt{\frac{2.25}{1}} = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2} \Rightarrow \theta_4 = \boxed{30^\circ = \theta_i}$$

(b) Lateral displacement:

$$d = t_1 \tan \theta_2 + t_2 \tan \theta_3 = 5 \tan 11.54^\circ + 5 \tan 19.47^\circ = 5(0.2041 + 0.3536) \text{ cm}$$

$$\boxed{d = 2.79 \text{ cm}}$$

HW3 P3: parallel-polarized wave through dielectric layers



Problem 4 — Dielectric Waveguide TE Mode Identification

20 pts

Setup: TE wave, $a = 5$ cm, $b = 3$ cm, $E_x = -36 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z)$ V/m.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$

TM mode $H_z = 0, E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) read m, n from arguments

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

TE mode allowed if at least one of $m, n \geq 1$. TM needs both $m, n \geq 1$.

CUTOFF FREQUENCY

f_{mn} m, n integers ≥ 0 (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ($a > b$)

Mode	f_c
TE ₁₀	$u_{p0}/(2a)$ (dominant)
TE ₀₁	$u_{p0}/(2b)$
TE ₂₀	$u_{p0}/a = 2f_{10}$
TE ₁₁ , TM ₁₁	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

 ϵ_r FROM DISPERSION

Given ω, β , mode (m, n)

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

$Z_{TE} > \eta$ always (denominator < 1); $Z_{TM} < \eta$.

(a) **Mode from the field expression.** Read $m\pi/a$ from the cos coefficient and $n\pi/b$ from the sin coefficient.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40 \cdot 0.05 = 2$$

$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100 \cdot 0.03 = 3$$

Mode: TE₂₃

(b) ϵ_r from dispersion, with $\omega = 2.4\pi \times 10^{10}$ rad/s, $\beta = 52.9\pi$ rad/m:

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] = \left(\frac{3 \times 10^8}{2.4\pi \times 10^{10}} \right)^2 [(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2]$$

$$\epsilon_r = 2.25$$

(c) Cutoff frequency for TE₂₃:

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s}$$

$$f_{23} = \frac{u_{p0}}{2} \sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{3}{b}\right)^2} = 10^8 \sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8 \sqrt{1600 + 10000}$$

$$f_{23} = 10.77 \text{ GHz}$$

(d) H_y from Z_{TE} : operating frequency $f = \omega/(2\pi) = 12$ GHz.

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} = \frac{377/\sqrt{2.25}}{\sqrt{1 - (10.77/12)^2}} = \frac{251.3}{\sqrt{1 - 0.805}} = \frac{251.3}{0.442} = 569.9 \text{ }\Omega$$

$$H_y = \frac{E_x}{Z_{TE}} = -0.063 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z) \text{ A/m}$$

Problem 5 — Hollow Guide, Multi-Mode Excitation & Travel Times 8 pts

Setup: Hollow guide ($\epsilon_r = 1$, $u_{p0} = c$), $a = 3$ cm, $b = 2$ cm. Pulse with 9.5 GHz carrier, guide length 100 m.

CUTOFF FREQUENCY

f_{mn} m, n integers ≥ 0 (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ($a > b$)

Mode	f_c
TE ₁₀	$u_{p0}/(2a)$ (dominant)
TE ₀₁	$u_{p0}/(2b)$
TE ₂₀	$u_{p0}/a = 2f_{10}$
TE ₁₁ , TM ₁₁	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \Rightarrow u_p u_g = u_{p0}^2$$

$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower $u_g \Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

Cutoff frequencies for the lowest modes:

$$f_{10} = \frac{c}{2a} = \frac{3 \times 10^8}{0.06} = 5 \text{ GHz}$$

$$f_{01} = \frac{c}{2b} = \frac{3 \times 10^8}{0.04} = 7.5 \text{ GHz}$$

$$f_{11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = (1.5 \times 10^8) \sqrt{(1/0.03)^2 + (1/0.02)^2} = 9.01 \text{ GHz}$$

$$f_{20} = \frac{c}{a} = 10 \text{ GHz} \quad (> 9.5, \text{ not excited})$$

Modes excited (carrier $\hat{f}_c$): TE₁₀, TE₀₁, TE₁₁, TM₁₁. TE₂₀ is just above cutoff so it does not propagate.

Group velocities: $u_g = c \sqrt{1 - (f_{mn}/f)^2}$ with $f = 9.5$ GHz.

Mode	f_c (GHz)	u_g/c	u_g (m/s)
TE ₁₀	5.00	$\sqrt{1 - (5/9.5)^2} = 0.85$	2.55×10^8
TE ₀₁	7.50	$\sqrt{1 - (7.5/9.5)^2} = 0.61$	1.84×10^8
TE ₁₁ , TM ₁₁	9.01	$\sqrt{1 - (9.01/9.5)^2} = 0.32$	0.95×10^8

Travel times: $t = L/u_g$ over 100 m.

Mode	t (μ s)
TE ₁₀	0.39
TE ₀₁	0.54
TE ₁₁ , TM ₁₁	1.05

Higher modes arrive later because they propagate closer to cutoff, so u_g drops.